Global Optimization via Softmin Energy Minimization

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Abstract

Non-convex optimization in high dimensions requires balancing gradient information with global exploration, a tension that Langevin dynamics and other gradient-informed methods address only partially. We introduce a gradient-based swarm method driven by a Softmin Energy interaction $J_\beta(\mathbf{x})$, a smooth approximation of the per-particle minimum that couples particles through their relative energies. Combining the resulting gradient flow with Brownian noise and an annealing schedule on β, we obtain a dynamics that retains the efficiency of gradient steps while inheriting the global-search behavior of swarm methods. We show, without any convexity assumption on the objective, that this interaction always partitions the ensemble into particles descending toward local minima and particles automatically repelled toward maxima, and that the same interaction caps the Freidlin–Wentzell quasipotential barrier separating adjacent basins at a level of order $1/\beta$, independently of the true barrier height, strictly below the barrier faced by an overdamped Langevin particle in the same landscape. Numerical experiments on double-well, quadruple-well, and Lennard-Jones cluster benchmarks confirm the analysis and show improved convergence over Simulated Annealing, Langevin dynamics, Gradient-Enhanced Particle Swarm Optimization, and Gradient-Aware Consensus Based Optimization (including their canonical gradient-free/unannealed variants).

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1 Introduction

Non-convex optimization problems are notoriously difficult, as the presence of multiple local minima can trap gradient-based algorithms. Existing approaches such as gradient descent perform well for convex functions with a single minimum but will likely converge to suboptimal local minima in more complex landscapes. This challenge is a central motivation in machine learning more broadly – non-convex losses in deep neural network training, and sharp local minima in reinforcement learning objectives, are two prominent settings where escaping suboptimal basins is a recognized bottleneck – though the benchmarks in this paper (Section 4) target structured, analytically tractable landscapes rather than these settings directly, so the theoretical and empirical claims below should be read as addressing the general non-convex escape problem, not as a validation on neural-network or RL training itself.

Preprint.

A classical strategy to address this issue is to add a noise term to the dynamics, allowing the system to escape local minima and leading to stochastic gradient–based methods, for example Langevin dynamics [10]. Stochastic gradient-based methods for optimization, while effective, typically exhibit poor transition times between local minima. The escape rate of a particle subject to Gaussian noise from a local minimum depends exponentially on the height of the potential barrier surrounding that minimum [5]. Methods that improve convergence rates over gradient-based algorithms often rely on additional assumptions about the objective function f. For example, if f admits a decomposition as a linear combination of simpler functions ψ_i, i.e., $f(x) = \sum_i \psi_i(x)$, then the Method of Moments [22] can be employed to converge to the global minimum. This type of decomposition is particularly useful in areas such as signal processing and machine learning, where complex functions are represented as sums of basis functions, such as Fourier series or wavelets [29]. In more general settings where such structure is not available, Simulated Annealing and its variants [7, 35] are widely used, leveraging a decaying temperature schedule to shift the dynamics from noisy exploration to refined local search.

A natural extension of stochastic gradient-based approaches is the use of multiple particles to explore the search space in parallel. For instance, when particles evolve independently under Langevin dynamics, the expected transition times between local minima decrease compared to single-particle dynamics purely due to statistical averaging over multiple independent trajectories. Moreover, allowing particles to cooperate rather than evolve independently can further improve exploration and convergence. The idea of particle cooperation is often central in Metaheuristic methods [4, 36]. Such methods provide a broad framework for global optimization and have demonstrated strong empirical performance, although establishing rigorous theoretical convergence guarantees is often challenging. Many such methods employ a particle-swarm-like paradigm [19, 27], in which multiple interacting particles explore the search space in parallel to locate global minimizers.

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SS: Check that this is factual and, eventually, cite some literature

SS: Maybe not worth citing, doesn't seem aligned with the comparative literature for this algorithm

SS: Cite IPLD literature

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The global optimization method proposed in this paper builds on these ideas by combining stochastic gradient techniques with cooperative multi-particle dynamics. We employ the well-known Softmin Energy functional J_β, which smoothly interpolates between the average and the hard minimum of the function values across particle positions. This functional acts as the source of interaction among particles, whose evolution is governed by its gradient, driving the system toward more effective exploration and optimization.

The Softmin function provides a natural way to interpolate between uniform weighting and hard minima via temperature $\beta > 0$. A common use of the Softmin function is to define the Gibbs distribution, where the probability of sampling index i out of values $v_1, \dots, v_n$ is given by:

$$\mathbb{P}[i] = \frac{\exp(-\beta v_i)}{\sum_{j=1}^n \exp(-\beta v_j)},$$

with $\beta \to \infty$ concentrating on $\min_i v_i$ and $\beta \to 0$ uniform [6]. This underlies stochastic optimization [1, 31, 16, 15] and exploration in Reinforcement Learning [24, 18].

Our proposed method is a stochastic gradient-based particle swarm approach that leverages a Softmin-type energy functional to effectively incorporate gradient information from the particles. Consider a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$. The objective is to approximately solve the global minimization problem $\min_{x \in \mathbb{R}^d} f(x)$. Given a population of n particles $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \in \mathbb{R}^d$ for all $i \in [n]$, the Softmin Energy is defined as:

$$J_\beta(\mathbf{x}) = \sum_{i=1}^n \frac{\exp(-\beta f(x_i))}{\sum_{j=1}^n \exp(-\beta f(x_j))} f(x_i), \tag{1}$$

which provides a smooth approximation of the empirical minimum among the $f(x_i)$. The parameter $\beta > 0$ controls the approximation's sharpness: larger values yield a closer approximation to the hard minimum $\min_i f(x_i)$, while $\beta \to 0$ recovers the average $\frac{1}{n}\sum_{i=1}^n f(x_i)$.

To minimize J_β we consider the stochastic gradient flow in the particle space $\mathbb{R}^{nd}$:

$$\mathrm{d}\mathbf{x}_t = -n\nabla J_{\beta_t}(\mathbf{x}_t)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t, \tag{2}$$

where $(B_t)_{t\geq 0}$ represents a standard nd-dimensional Brownian motion and $\sigma > 0$ introduces stochasticity to promote exploration and escape from local minima.

1.1 Contribution

The main novelty of this work is the rigorous analysis of stochastic gradient-based dynamics driven by the Softmin Energy functional, $J_\beta(\mathbf{x})$. In this paper, we prove that the resulting inter-particle

interaction mechanism provides strict theoretical advantages over standard gradient-based and non-interacting methods for non-convex global optimization.

SS: Maybe too strong as a claim?

Dynamic Ensemble Partitioning and Accelerated Convergence The interaction induced by ∇J_β naturally partitions the particle ensemble via an effective learning rate, $A_k(\mathbf{x})$. At every stable equilibrium, and without any convexity assumption on f, particles below the Softmin threshold ($A_k(\mathbf{x}) > 0$) sit at local minima, while trapped, high-energy particles ($A_k(\mathbf{x}) < 0$) are automatically repelled to hill-climb and explore, with at least one particle always at a local minimum (Theorem 2.1). Along the trajectory, the sign of $A_k(\mathbf{x})$ is not conserved for any single particle; instead, we prove that the running best objective value across the whole ensemble decreases at least as fast as ordinary gradient descent whenever f satisfies a Polyak–Łojasiewicz condition, for any – possibly time-varying – inverse-temperature schedule (Theorem 2.4).

SS: Referential to old version. Moreover, the PL condition is only necessar to have convergence guarantee to the global optimum

Provable Barrier Flattening We demonstrate that the Softmin dynamics do not merely reduce effective potential barriers, but analytically cap them. By integrating the effective drift along an explicit escape channel, we prove that arbitrarily high physical barriers separating adjacent basins are capped, at the level of the Freidlin–Wentzell quasipotential, to a constant of order $1/\beta$, uniformly in the true barrier height (Theorem 3.5). This bound holds unconditionally on the quasipotential itself, strictly below the exact small-noise exit exponent of an overdamped Langevin particle in the same landscape; turning it into an exponentially cheaper guarantee on the expected transition time additionally requires the metastable-domain hypotheses made precise in Corollary 3.6. This structural landscape modulation is a direct consequence of the continuous Softmin repulsion, distinguishing our approach from generic stochastic swarm methods and Simulated Annealing.

1.2 Related work

Metaheuristic optimization methods Metaheuristic optimization methods include Consensus-Based Optimization (CBO) [8, 13, 25, 9], Genetic Algorithms [17], Ant Colony Optimization [12], and Particle Swarm Optimization (PSO) [20, 23]. Traditional particle-swarm approaches typically operate without explicit gradient information, relying instead on position-based updates to coordinate particles toward promising regions. When reliable gradient information is available, leveraging it enables convergence rates comparable to standard stochastic gradient descent, which gradient-agnostic metaheuristics cannot achieve.

Recent gradient-based swarm methods [28, 7, 32, 9, 23], have made progress by incorporating gradient information into multi-particle dynamics. However, our approach differs fundamentally in its mechanism for inducing exploration and escaping barriers. Rather than consensus-based averaging or general gradient-weighted potentials, we employ the Softmin Energy functional to explicitly create a minimizing/maximizing partition of the particle ensemble: particles below the Softmin threshold are drawn toward local minima via gradient descent, while particles above the threshold are repelled toward maxima. This automatic role assignment (Theorem 2.1) simultaneously reduces effective potential barriers (Theorem 3.5) and forces suboptimal particles into exploratory regimes without requiring manual basin-switching heuristics.

To the best of our knowledge, no other metaheuristic employs the Softmin Energy functional, computed as the softmin of function values at particle positions, as the sole source of inter-particle interaction. This design enables provable escape from suboptimal local minima with a barrier bounded independently of the true barrier height (Theorem 3.5, Section 3), together with a deterministic guarantee that the ensemble's best value never lags ordinary gradient flow under a Polyak–Łojasiewicz condition (Theorem 2.4, Section 2.2), as validated experimentally on benchmark landscapes where barriers dominate the hardness.

Langevin-based sampling and interacting-particle methods Stochastic Gradient Langevin Dynamics (SGLD) [34] and its non-asymptotic analysis for non-convex optimization [26] study a single diffusion whose drift is the gradient of the objective itself, with guarantees governed by the spectral gap of the associated Gibbs measure; when multiple particles are used, interaction typically enters only through averaging over independent runs. Interacting Particle Langevin Dynamics (IPLD) [2, 11] instead couples an ensemble of particles through their empirical measure, using this mean-field interaction to build unbiased or stabilized estimators of the marginal likelihood gradient in latent-variable models. Our construction is closer in spirit to IPLD, in that it also relies on genuine

ensemble-level coupling rather than independent copies, but the interaction itself is different in kind: instead of pooling particles through an empirical average, the Softmin Energy J_β couples particles through their relative energies, producing a state-dependent, sign-changing effective learning rate $A_k(\mathbf{x})$ that deterministically assigns each particle a minimizing or maximizing role (Theorem 2.1). It is this competitive, temperature-sharpened coupling, rather than mean-field averaging, that produces the barrier-capping mechanism of Theorem 3.5, and it is aimed directly at global optimization of f rather than at marginal-likelihood estimation.

Connection to Statistical Physics and Functional Design Our approach shares conceptual roots with population-based statistical physics methods such as Population Annealing (PA) [33], which also employs a Gibbs-weighted ensemble to locate global ground states. However, while PA relies on discrete demographic selection, duplicating low-energy particles and eliminating high-energy ones prior to local Markov Chain Monte Carlo steps, our method maintains a fixed population size and drives exploration through continuous dynamical repulsion. This active exploration mechanism differentiates our Softmin Energy functional from the standard log-sum-exp (or free-energy) functional, $F_\beta(\mathbf{x}) = -\beta^{-1}\log\sum e^{-\beta f(x_i)}$. The gradient of the log-sum-exp formulation, $\nabla_{x_k} F_\beta(\mathbf{x}) = w_k \nabla f(x_k)$, strictly preserves gradient descent, causing sub-optimal particles with small weights to simply stagnate. In contrast, the gradient of our proposed Softmin functional, $J_\beta(\mathbf{x}) = \sum w_i f(x_i)$, natively incorporates the sign-changing effective learning rate, $A_k(\mathbf{x})$. Consequently, rather than discarding trapped particles, the geometry of the Softmin functional actively destabilizes their local minima, forcing high-energy particles to hill-climb and escape via the energy barrier-flattening mechanism formalized in our theoretical analysis.

Softmin Energy Theoretical work on the Gibbs Softmin (or Boltzmann Softmin) has primarily focused on its properties as a smooth approximation to the minimum operator [6], convergence of softmax policy gradients in reinforcement learning [21], and its role in exploration–exploitation trade-offs via temperature scheduling [30]. To our knowledge, this is the first work employing J_β as an optimization objective for non-convex global search. The effective barrier reduction via particle cooperation, analyzed in Theorem 3.5, is previously unstudied.

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1.3 Structure of the paper

In Section 2, we analyze the deterministic gradient flow dynamics obtained by setting $\sigma = 0$ in (2) in detail. Section 2.1 characterizes the stable equilibria of this flow with no convexity assumption on f: every particle sits at a critical point of f, the sign of its effective learning rate fixes whether that critical point is a minimum or a maximum, and at least one particle always resides at a local minimum. Section 2.2 then shows that, whenever f satisfies a Polyak–Łojasiewicz condition, the best objective value across the ensemble decreases at least as fast as ordinary gradient descent, for any inverse-temperature schedule.

In Section 3, we examine the noisy dynamics (2) at fixed inverse temperature, showing that the Softmin interaction caps the Freidlin–Wentzell quasipotential barrier separating adjacent basins at a level of order $1/\beta$, independently of the true barrier height – strictly below the exact small-noise exit exponent of an overdamped Langevin particle in the same landscape.

In Section 4, we conduct numerical experiments to validate the method's effectiveness in escaping local minima and identifying global optima. Comparisons with Simulated Annealing, Gradient-Aware CBO [9], and Gradient-Enhanced PSO [23] show better performance, particularly when potential barriers exceed a critical threshold.

The proofs of lemmas and theorems throughout the paper are deferred to Appendix B. Details on experimental settings can be found in Appendix C.

2 Gradient Flow dynamics for Softmin Energy

This section focuses on the deterministic gradient flow dynamics

$$\mathrm{d}\mathbf{x}_t = -n\nabla J_\beta(\mathbf{x}_t)\mathrm{d}t, \tag{3}$$

where $\mathbf{x}_t = (x_1(t), \ldots, x_n(t))$ denotes the state of the particle system and the Softmin Energy $J_\beta(\mathbf{x})$ is defined in (1). The scaling factor n is fixed by the requirement that the dynamics reduce, in the high-

temperature limit $\beta \to 0$, to n independent gradient (and, once noise is added, Langevin) particles: as shown below, every Gibbs weight $w_k \to 1/n$ and every effective learning rate $A_k(\mathbf{x}) \to 1$ in this limit, so (3) recovers $\dot{x}_k = -\nabla f(x_k)$ componentwise – the consistency check underlying the comparison with independent Langevin dynamics in Remark 3.9. It also ensures that the subsequent derivatives and governing equations remain well-posed in the mean-field limit $n \to \infty$. This continuous-time system performs gradient descent on the Softmin landscape, which is formulated such that its minimization inherently drives the ensemble toward the minima of the objective function f.

Subsequent analysis characterizes the stable equilibria of this flow without any convexity assumption on f: at every such equilibrium, each particle sits at a critical point of f, the sign of its effective learning rate $A_k(\bar{\mathbf{x}})$ fixes whether that critical point is a minimum or a maximum, and at least one particle resides at a local minimum. Separately, we show that the best objective value across the ensemble decreases at least as fast as ordinary gradient descent whenever f satisfies a Polyak–Łojasiewicz condition, for any (possibly time-varying) inverse temperature schedule.

To formally analyze these dynamics, it is necessary to first evaluate the gradient of nJ_β with respect to the individual particle coordinates. Let $w_k = e^{-\beta f(x_k)}(\sum_j e^{-\beta f(x_j)})^{-1}$ denote the Gibbs weight of particle x_k, and define the dynamic pre-factor $A_k(\mathbf{x}) = nw_k\,[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))]$. The gradient of $nJ_\beta(\mathbf{x})$ with respect to particle x_k is computed in Lemma B.1 and is given by:

$$\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_k(\mathbf{x})\nabla f(x_k), \tag{4}$$

so that (3) reads componentwise as $\dot{x}_k = -A_k(\mathbf{x})\nabla f(x_k)$. The exact second-order partial derivatives and corresponding Hessian blocks, which are required for the subsequent local stability analysis, are deferred to Appendix B.1.

This formulation reveals a critical property of the Softmin gradient. The dynamic pre-factor $A_k(\mathbf{x}) = nw_k\beta\,[1/\beta - f(x_k) + J_\beta(\mathbf{x})]$ dictates the effective direction of the gradient flow for each particle x_k, acting as a state-dependent learning rate. According to the dynamics in (3), if a particle evaluates to a landscape value $f(x_k) < J_\beta(\mathbf{x}) + 1/\beta$, the pre-factor $A_k(\mathbf{x})$ is strictly positive, and the particle performs standard gradient descent on f; the particle is then said to be *minimizing*. Conversely, if $f(x_k) > J_\beta(\mathbf{x}) + 1/\beta$, the pre-factor becomes negative, effectively reversing the flow, and the particle is *maximizing*, executing gradient ascent on f. This mechanism theoretically guarantees that particles trapped in sub-optimal local minima are actively repelled and forced to explore the landscape, whereas particles at comparatively lower energy states continue to minimize f.

Two elementary facts drive the equilibrium analysis that follows. First, summing the pre-factor over all particles and using $\sum_k w_k = 1$ and $\sum_k w_k f(x_k) = J_\beta(\mathbf{x})$,

$$\sum_{k=1}^{n} A_k(\mathbf{x}) = n \qquad \text{for every } \mathbf{x} \in \mathbb{R}^{nd}, \tag{5}$$

so at least one coefficient is always strictly positive. Second, at any configuration where every particle sits at a critical point of f, the mixed second partial derivatives of J_β vanish identically, since each mixed Hessian block is an outer product carrying a factor $\nabla f(x_k)$ (Lemma B.1); the Hessian of nJ_β is then block diagonal.

2.1 Stationary points of the dynamics

This subsection characterizes the stable equilibria of the gradient flow (3), with *no convexity assumption whatsoever* on the objective function f. Theorem 2.1 shows that at any such equilibrium every particle sits at a critical point of f with a strictly nonzero pre-factor, the sign of that pre-factor fixes whether the critical point is a minimum or a maximum, and at least one particle always resides at a local minimum. Consequently, the Softmin formulation guarantees both the exploitation of deep minima and the systematic destabilization of higher-energy critical points, without requiring any local geometric hypothesis on f.

Theorem 2.1. *Let $f \in C^2(\mathbb{R}^d)$, let $\beta > 0$, and let $\bar{\mathbf{x}} = (\bar{x}_1, \ldots, \bar{x}_n)$ be a locally asymptotically stable equilibrium of* (3)*. Then, for every $k \in \{1, \ldots, n\}$, evaluated at $\bar{\mathbf{x}}$:*

1. $\nabla f(\bar{x}_k) = 0$;

2. $A_k(\bar{\mathbf{x}}) \neq 0$*;*

3. *the sign of* $A_k(\bar{\mathbf{x}})$ *fixes the type of* $\bar{x}_k$*:*
   $A_k(\bar{\mathbf{x}}) > 0 \implies \bar{x}_k$ *is a local minimum of* f*,* $\quad A_k(\bar{\mathbf{x}}) < 0 \implies \bar{x}_k$ *is a local maximum of* f*,*
   with $A_k(\bar{\mathbf{x}}) > 0 \iff f(\bar{x}_k) < J_\beta(\bar{\mathbf{x}}) + 1/\beta$*.*

Moreover, by (5)*, at least one particle sits at a local minimum of* f*.*

Proof. The rigorous derivation of these stability conditions is deferred to Appendix B. □

Corollary 2.2 (Sufficient condition for stability)**.** *Let* $\bar{\mathbf{x}} = (\bar{x}_1, \dots, \bar{x}_n)$ *satisfy* $\nabla f(\bar{x}_k) = 0$ *for every* k*, and assume that for each* k *either*

$$A_k(\bar{\mathbf{x}}) > 0 \text{ and } \nabla^2 f(\bar{x}_k) \succ 0, \qquad \text{or} \qquad A_k(\bar{\mathbf{x}}) < 0 \text{ and } \nabla^2 f(\bar{x}_k) \prec 0,$$

evaluated at $\bar{\mathbf{x}}$*. Then* $\nabla^2(nJ_\beta)(\bar{\mathbf{x}}) \succ 0$*; in particular* $\bar{\mathbf{x}}$ *is a strict local minimum of* nJ_β *and a locally asymptotically stable equilibrium of* (3)*.*

Proof. By Lemma B.1 the Hessian at a fully critical configuration is block diagonal with blocks $A_k(\bar{\mathbf{x}})\nabla^2 f(\bar{x}_k)$, and each hypothesis makes the corresponding block positive definite. □

The deterministic flow can therefore become trapped at any equilibrium consisting of particles at local minima and local maxima of f, not only at full consensus on a single (possibly suboptimal) minimum; the noisy dynamics analyzed next provide a mechanism for the ensemble to escape such traps.

2.2 Descent of the ensemble minimum

The instantaneous sign of the pre-factor $A_k(\mathbf{x}(t))$ is not conserved for a fixed particle, since it depends on every coordinate through J_β. The invariant object is instead the running best value across the ensemble,

$$m(t) := \min_{1 \le k \le n} f(x_k(t)),$$

which we now control for a general, possibly time-dependent, inverse-temperature schedule $\beta_t > 0$. Write $A_k(\mathbf{x}(t))$ for the pre-factor evaluated at inverse temperature β_t and configuration $\mathbf{x}(t)$.

Definition 2.3 (Polyak–Łojasiewicz condition)**.** A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ satisfies the *Polyak–Łojasiewicz (PL) condition* on a set $S \subseteq \mathbb{R}^d$ with constant $\mu > 0$ if

$$\tfrac{1}{2}\|\nabla f(x)\|^2 \ge \mu\big(f(x) - f_\star\big), \qquad f_\star := \inf f,$$

for every $x \in S$.

Theorem 2.4 (Descent of the ensemble minimum)**.** *Let* $f \in C^1(\mathbb{R}^d)$ *have locally Lipschitz gradient, let* $\beta_t > 0$ *be measurable, and let* $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))$ *be a global solution of*

$$\dot{x}_k(t) = -A_k(\mathbf{x}(t))\,\nabla f(x_k(t)), \qquad k = 1, \dots, n.$$

Suppose f *satisfies the Polyak–Łojasiewicz condition (Definition 2.3) with constant* $\mu > 0$ *on the sublevel set* $S_0 := \{x : f(x) \le m(0)\}$*. Then* m *is nonincreasing and*

$$m(t) - f_\star \le e^{-2\mu t}\big(m(0) - f_\star\big), \qquad t \ge 0.$$

Proof. The proof is deferred to Appendix B. □

The estimate compares the best value in the ensemble against ordinary gradient flow; it does not assign a permanent role to any designated particle, and it holds for any positive schedule β_t, in every dimension. Note that the Polyak–Łojasiewicz hypothesis is imposed on the sublevel set S_0 itself: it holds automatically whenever S_0 contains no suboptimal critical point of f, and Theorem 2.4 should be read as a no-slowdown guarantee for the ensemble's leading particle in that regime, complementary to the basin-escape mechanism of Theorem 3.5, rather than as a global convergence-rate result.

Remark 2.5 (Global existence). Theorem 2.4 presupposes a global solution. Since m is nonincreasing, every particle attaining the running minimum stays in the compact sublevel set S_0 when f is coercive; the theorem controls these particles only. Maximizing particles ($A_k(\mathbf{x}(t)) < 0$) are damped by an exponentially small Gibbs weight, but their trajectories are not controlled by this estimate, and global existence must be assumed or established separately.

3 Barrier Flattening and Exit Asymptotics

This section specializes (2) to a fixed inverse temperature, $\beta_t \equiv \beta$, with the noise term $\sqrt{2\sigma}\mathrm{d}B_t$ introducing isotropic nd-dimensional Brownian noise. Componentwise,

$$\mathrm{d}x_k(t) = -A_k(\mathbf{x}_t)\nabla f(x_k(t))\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t^k, \tag{6}$$

so $A_k(\mathbf{x}_t)$ acts as a state-dependent, sign-changing effective learning rate for particle x_k, as established in Section 2. The rigorous exit analysis below is carried out for *fixed* β and physical dimension $d = 1$; the time-inhomogeneous (annealed) case $\beta = \beta_t$ requires a separate treatment and is used only heuristically in Section 4. The ensemble-descent estimate of Section 2.2, by contrast, holds for any positive schedule and in every dimension.

3.1 One-dimensional basin exit and post-saddle passage

Fix $n \geq 2$, $\beta > 0$, and $d = 1$. Consider three consecutive nondegenerate critical points of f,

$$z_- < s < z_+,$$

with $z_\pm$ local minima and s the separating local maximum, and assume

$$f'(x) > 0 \text{ on } (z_-, s), \qquad f'(x) < 0 \text{ on } (s, z_+). \tag{7}$$

Set the relative energy and the barrier height

$$r(x) := f(x) - f(z_-), \qquad h := r(s) > 0, \qquad \rho_+ := r(z_+) < h. \tag{8}$$

No sign is imposed on ρ_+; the case of interest is $\rho_+ < 0$, in which z_+ is a strictly deeper minimum than z_-.

Assumption 3.1. $f \in C^2(\mathbb{R})$; $z_- < s < z_+$ are nondegenerate critical points of f (i.e. $f''(z_-) > 0$, $f''(s) < 0$, $f''(z_+) > 0$) satisfying the monotonicity condition (7); and $n \geq 2$, $\beta > 0$, $d = 1$.

Remark 3.2 (Metastability of the initial state). At the consensus configuration $\bar{\mathbf{x}} = (z_-, \ldots, z_-)$, every Gibbs weight equals $1/n$, so $A_k(\bar{\mathbf{x}}) \equiv 1$ for every k. By Corollary 2.2, using $f''(z_-) > 0$, $\bar{\mathbf{x}}$ is therefore a strict local minimum of nJ_β and a locally asymptotically stable equilibrium of (3), i.e. genuinely metastable for the noisy dynamics (2) at small σ – and, once the channel Γ below identifies $\bar{\mathbf{x}}$ with $\Gamma(z_-)$, this is exactly the metastable state the exit-time analysis of this section departs from.

We rely on the Freidlin–Wentzell large deviation theory for exit times of small-noise diffusions from metastable domains, summarized in Appendix A: under the standard hypotheses of that theory, the exit time τ_D^σ from a metastable domain D satisfies $\lim_{\sigma\downarrow 0} \sigma \log \mathbb{E}_{\mathbf{x}} \tau_D^\sigma = V_\beta(\mathbf{x}, \partial D)$, where V_β is the Freidlin–Wentzell quasipotential associated with (2) at fixed β.

To build an admissible escape path, freeze $n-1$ particles at z_- and let one particle move:

$$\Gamma(x) := (x, z_-, \ldots, z_-). \tag{9}$$

Factoring $e^{-\beta f(z_-)}$ out of the Gibbs weights gives $J_\beta(\Gamma(x)) = f(z_-) + u(r(x))/n$, that is, the exact identity

$$nJ_\beta(\Gamma(x)) = nf(z_-) + u(r(x)), \qquad u(r) := \frac{n\, r\, e^{-\beta r}}{n - 1 + e^{-\beta r}}, \quad r \in \mathbb{R}, \tag{10}$$

where u is the *reduced potential*. The additive constant $nf(z_-)$ does not enter $\nabla(nJ_\beta)$, and both the action and the quasipotential depend on nJ_β only through its increments. Note that u is defined on all of $\mathbb{R}$, since $r(x)$ is negative wherever f drops below $f(z_-)$.

Lemma 3.3 (Channel invariance and reduced gradient flow)**.** *The set $\{\Gamma(x) : x \in \mathbb{R}\}$ is invariant under* (3)*, since $f'(z_-) = 0$ forces the frozen coordinates' drift to vanish identically. On this invariant manifold, the dynamics reduces to the one-dimensional gradient flow $\dot{x} = -W'(x)$ of the reduced potential $W(x) := u(r(x))$ of* (10)*.*

Proof. Since z_- is a critical point of f, every frozen coordinate satisfies $\dot{x}_j = -A_j(\Gamma(x))f'(z_-) = 0$ by (3), so $\{\Gamma(x) : x \in \mathbb{R}\}$ is invariant. Differentiating (10) along the channel gives $W'(x) = \frac{d}{dx}\big[nJ_\beta(\Gamma(x))\big] = u'(r(x))r'(x) = u'(r(x))f'(x)$; applying the chain rule directly to nJ_β and (4) gives the same derivative as $A_1(\Gamma(x))f'(x)$. Hence $W'(x) = A_1(\Gamma(x))f'(x)$, matching the active coordinate's drift $-A_1(\Gamma(x))f'(x)$ in (3), and in particular reproving $A_1(\Gamma(x)) = u'(r(x))$, made explicit in closed form in Lemma 3.4 below. □

The reduced potential also records the sign of the drift of the active particle, and this is what fixes the geometry of the channel.

Lemma 3.4 (Channel drift and shape of the reduced potential)**.** *Let $A_1(\Gamma(x))$ denote the pre-factor of the active particle along the channel. Then, for every $x \in \mathbb{R}$,*

$$A_1(\Gamma(x)) = u'(r(x)) = \frac{n\, e^{-\beta r(x)}\, q(r(x))}{\big(n-1+e^{-\beta r(x)}\big)^2}, \qquad q(r) := (n-1)(1-\beta r) + e^{-\beta r}. \tag{11}$$

The function q is strictly decreasing on $\mathbb{R}$ and has a unique zero r_c, which satisfies

$$\frac{1}{\beta} < r_c < \frac{n}{(n-1)\beta}. \tag{12}$$

Consequently u is strictly increasing on $(-\infty, r_c]$ and strictly decreasing on $[r_c, \infty)$, and r_c is exactly the relative-energy level at which the active particle switches from minimizing to maximizing, i.e. the level at which $f(x) - J_\beta(\Gamma(x)) = 1/\beta$.

Proof. See Appendix B. □

We abbreviate the two relevant channel heights by

$$u_c := u(r_c) = \max_{r\in\mathbb{R}} u(r), \qquad u_h := \max_{0\le r\le h} u(r) = u\big(\min\{h, r_c\}\big). \tag{13}$$

Theorem 3.5 (One-dimensional basin exit and post-saddle passage)**.** *Under Assumption 3.1, the following hold.*

1. ***(Uniform barrier cap.)*** *The channel barrier is bounded independently of the physical barrier h and of f:*

$$u_c = \max_{r\in\mathbb{R}} u(r) < \frac{n}{e(n-1)\beta}. \tag{14}$$

In particular both the barrier u_c and, by (12)*, the switching level r_c are of order β^{-1}, uniformly in h.*

2. ***(Initial basin exit: quasipotential bound.)*** *The minimum action among channel paths joining $\Gamma(z_-)$ to $\Gamma(s)$ equals u_h. Consequently*

$$V_\beta\big(\Gamma(z_-), \Gamma(s)\big) \le u_h < \min\{h, r_c\} \le h. \tag{15}$$

3. ***(Post-saddle well and passage.)*** *Assume*

$$r_c < h. \tag{16}$$

Then $\Gamma(s)$ is a strict local minimum of nJ_β, with channel barrier $u_c - u(h)$ on the z_- side (realized at x_m). The behavior on the z_+ side depends on where ρ_+ falls relative to r_c:

- *If $\rho_+ < r_c$, there is a unique $\xi_+ \in (s, z_+)$ with $r(\xi_+) = r_c$; the channel well at $\Gamma(s)$ has matching barrier height $u_c - u(h)$ on this side as well, the minimum action among channel paths joining $\Gamma(s)$ to $\Gamma(\xi_+)$ equals $u_c - u(h)$, and therefore*

$$V_\beta\big(\Gamma(s), \Gamma(\xi_+)\big) \le u_c - u(h) < u_c; \tag{17}$$

  beyond ξ_+ the deterministic channel dynamics drive the active particle to z_+.
- *If instead $\rho_+ \ge r_c$, then $r(x) \in [r_c, h)$ throughout (s, z_+); since u is strictly decreasing on $[r_c, \infty)$ and r is strictly decreasing on (s, z_+), the channel potential $u(r(x))$ increases monotonically from $u(h)$ at s to $u(\rho_+)$ at z_+, with no interior peak: on this side $\Gamma(s)$ is not separated from $\Gamma(z_+)$ by any channel barrier, and no point ξ_+ exists.*

Consequently, when $\rho_+ < r_c$, the two-stage passage from $\Gamma(z_-)$ through $\Gamma(s)$ to $\Gamma(\xi_+)$ satisfies

$$V_\beta\big(\Gamma(z_-), \Gamma(\xi_+)\big) \;\le\; u_h + \big(u_c - u(h)\big) \;\le\; 2u_c - u(h) \;<\; \frac{2n}{e(n-1)\beta}, \tag{18}$$

by (15)*,* (17)*, and $u_h \le u_c$: the composite barrier remains of order β^{-1}, uniformly in h. The corresponding exit-time exponent for the two crossings is governed by the larger of the two, $\max\{u_h, u_c - u(h)\} \le u_c$, not by their sum. If instead $h \le r_c$, then $\Gamma(s)$ is not a local minimum of nJ_β and no post-saddle well forms.*

Proof. See Appendix B. □

The scope of Theorem 3.5 and its relation to the exact exit-time exponent is discussed in Remarks 3.8–3.10 below.

Corollary 3.6 (Exit-time consequence, conditional)**.** *Suppose in addition that z_- admits a metastable domain $D_- \subset \mathbb{R}^n$ for the flow* (3)*, in the sense that $\Gamma(z_-)$ is the unique attractor of D_- for the noiseless flow and the hypotheses of the Freidlin–Wentzell exit theorem (Appendix A) hold on D_-, with $\Gamma(x_m) \in \partial D_-$ realizing the minimal quasipotential to ∂D_-, where $x_m \in (z_-, s]$ is the unique point with $r(x_m) = \min\{h, r_c\}$. Then*

$$\lim_{\sigma \downarrow 0} \sigma \log \mathbb{E}_{\Gamma(z_-)} \tau^{\sigma}_{D_-} = V_\beta\big(\Gamma(z_-), \partial D_-\big) \le u_h < h.$$

We do not construct D_- or verify these hypotheses here; Theorem 3.5 supplies only the quasipotential bound (15)*, which holds unconditionally.*

Remark 3.7 (Low-barrier regime: acceleration without sign reversal). By (11), $A_1(\Gamma(x)) > 0$ precisely when $r(x) < r_c$. Hence $u_h = u(h)$, i.e. $\min\{h, r_c\} = h$, exactly in the regime $h \le r_c$, and there the drift of the active particle is minimizing along the entire crossing: the sign-reversal mechanism is never activated, the crossing is made entirely against the drift, and by Theorem 3.5(3) no Softmin-induced well appears at s. The acceleration is nonetheless strict, since $u(r)/r = ne^{-\beta r}/(n - 1 + e^{-\beta r}) < 1$ for $r > 0$ gives $u(h) < h$. In this regime the gain comes purely from the Gibbs-weight damping of the leading particle's contribution to nJ_β, not from a sign reversal. By (12) the regime is $h \lesssim \beta^{-1}$; the sign-reversal mechanism is what handles barriers beyond that scale.

Remark 3.8 (Interpretation and scope). Part 1 shows that, along this channel, an arbitrarily high physical barrier h is flattened to a level of order β^{-1}: as h grows the active particle receives exponentially small Gibbs weight, so the barrier saturates rather than growing with h. Part 2 gives a strict full-system *upper* bound on the exit exponent through an explicit admissible path; because V_β is an infimum over all paths in $\mathbb{R}^n$, the channel construction can only over-estimate it. It is a bound on the Freidlin–Wentzell quasipotential itself, established by exhibiting an admissible path; it does not yet assert anything about $\mathbb{E}\tau^\sigma$, for which see Corollary 3.6. Part 3 addresses the post-saddle obstruction: a high barrier turns the physical maximum s into a Softmin-induced well. When $\rho_+ < r_c$ this well has equal-height barriers, with depth $u_c - u(h)$ strictly below the initial barrier u_c, so it is not a worse obstruction than the one already crossed, giving the composite two-stage bound (18); when $\rho_+ \ge r_c$ there is no obstruction on the z_+ side at all, which is only more favorable. The theorem does *not* identify the exact transition exponent between the two basins, which would require ruling out returns to the initial basin and competing collective paths in $\mathbb{R}^n$.

Remark 3.9 (Comparison with independent Langevin dynamics). For a single overdamped Langevin particle in the potential f started at z_-, the classical small-noise exit exponent from the basin of z_- is the physical barrier $h = f(s) - f(z_-)$. For n *independent* copies, the first of the n exit times has the same exponential scale h: taking a minimum over n i.i.d. variables with common exponential rate h/σ changes only the multiplicative prefactor, by a factor $1/n$, not the exponent, as $\sigma \downarrow 0$.

By (15), the channel construction gives $V_\beta(\Gamma(z_-), \Gamma(s)) \le u_h < h$ for the joint Softmin diffusion, in *every* regime of h (see Remark 3.7). This is a comparison of an *upper bound on a quasipotential* for the interacting system against the *exact* exponent h of the noninteracting one; it shows the admissible channel gives a strictly lower quasipotential bound than independent escape, not that the true joint exit exponent is smaller. The stronger, exit-*time* statement that the interaction cannot slow escape holds only under the hypotheses of Corollary 3.6, which are not verified in this paper.

Remark 3.10 (Higher dimensions, temperature, and scale). The channel identity of Part 1 extends algebraically to $d > 1$: at a critical point s of f the active Hessian block is $u'(h)\nabla^2 f(s)$, so if s is an index-one saddle and $u'(h) < 0$ the induced local minimum is a genuinely one-dimensional phenomenon, and transverse unstable directions may open cheaper escape routes. Regarding temperature, the fixed-β bound (14) decreases like β^{-1}, favoring large β for barrier reduction; but large β also suppresses the drift of non-leading particles through their Gibbs weights, since a particle trailing the ensemble minimum by a gap Δ is weighted by $e^{-\beta\Delta}$. Both limits degrade the mechanism: as $\beta \to \infty$ the ensemble degenerates to leader-only drift, while as $\beta \to 0$ the Softmin coupling degenerates toward independent noisy gradient descent and (14)'s bound itself diverges. A decaying (or, symmetrically, an increasing) schedule β_t is a natural candidate for balancing these two effects over time; however, we found empirically that neither improves on a well-chosen *fixed* β on the benchmarks considered here (Section 4) – both a geometric decay from a large starting β and a

geometric growth from a small one consistently reach the target *more slowly* than a fixed β tuned once via the diagnostic of Appendix C, with the gap widening at harder barrier heights, and an aggressive growth rate can drive β to numerical overflow before the schedule ever reaches a useful range. We report this as a negative empirical result rather than infer a general claim about annealing from it: a rigorous large-deviation treatment of β_t, and of whether some other schedule shape or landscape can realize the exploration/refinement trade-off this remark motivates, remain open.

4 Numerical Experiments

We evaluate the capacity of Softmin dynamics to escape local minima and outrun sub-optimal basins by comparing fixed-β ($\beta = 10$) Softmin against seven baselines: Simulated Annealing (SA) [16]; overdamped Langevin dynamics [34], a fixed-noise, zero-hyperparameter control; Gradient-Aware Consensus-Based Optimization (CBO) [9], with and without upward β-annealing; Gradient-Enhanced Particle Swarm Optimization (PSO) [23]; and the canonical, gradient-free formulations of both CBO [8, 25] and PSO [20]. The continuous dynamics are discretized using the Euler-Maruyama scheme, and entry time is measured against a value-space target – a fixed fraction of each benchmark's own depth gap for the wells, or a fixed tolerance above the literature global minimum for Lennard-Jones – rather than a coordinate-space box, so difficulty scaling is not conflated with an arbitrary geometric artifact. Landscape formulations, hyperparameters, initialization bounds, hardware specifications, and real per-particle gradient/objective evaluation counts and wall-clock time are detailed in Appendix C; the source code is available at [3].

We first verify the large deviation scaling of Theorem 3.5 on a 1D double well ($\sigma = 0.05$); results are in Appendix C.4, alongside the ascent-descent mechanism ablation, the per-particle prefactor illustration, and the full cost comparison.

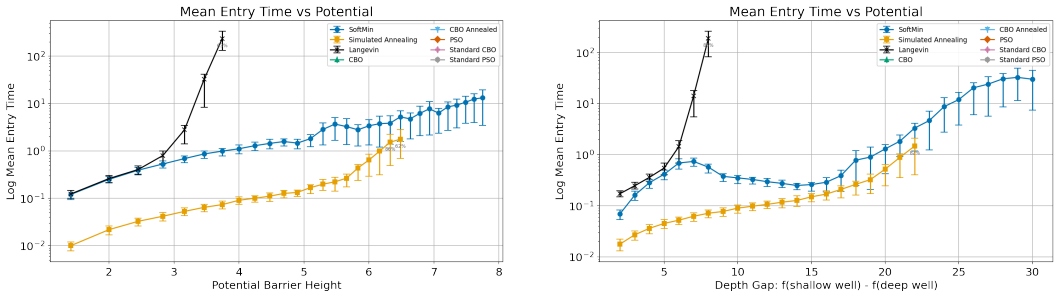


Figure 1: Mean first-entry time ($\pm$ IQR) into a value-space target, $\sigma = 1$. Left: 1D double well vs. barrier height h. Right: 2D quadruple well vs. depth gap between the shallow and deep wells. Points below a 90% reached-fraction are annotated with the exact fraction; CBO and PSO – gradient-aware and canonical alike – never reach the target at any swept difficulty on either benchmark and so never appear. The same 100 seeds (`seed=0,...,99`, see §C.3) are reused at every sweep point rather than redrawn independently, so the quartile bands reflect within-seed variability, not independent per-point samples.

Figure 1 (Left) reports 1D performance across barrier heights h. Softmin reaches the target at every h tested, in all 100 seeds, with mean entry time growing gently across the swept range. SA is consistently faster than Softmin wherever it still succeeds – at the mid-sweep point $\delta = 15$, for instance, SA reaches the target in a mean 0.22 against Softmin's 3.67 – but its reached-fraction degrades with h and collapses entirely by $\delta = 30$ (from 100% at $\delta = 15$ to 1% at $\delta = 25$ to 0%); Langevin fails even earlier, already at 0% by $\delta = 15$ having succeeded at $\delta = 5$. CBO and PSO, gradient-aware and canonical alike, never reach the target at any barrier height we tested. Softmin is therefore not the fastest method in the regime where SA still works, but it is the only one that keeps working as the barrier grows.

Figure 1 (Right) extends this to the corrected 2D quadruple well, whose `particle_init`/target now sit at genuinely different depths – a depth gap growing linearly in A (Appendix C.3, Table 1) – rather than the exact mirror pair used previously, which had no suboptimal basin to escape at

all. The same pattern holds and sharpens: Softmin reaches the target at every depth gap tested (2 through 30, 100% of seeds throughout), while SA is faster where it still succeeds (e.g. 0.15 mean entry time at $A = 15$ against Softmin's 0.26) but its reached-fraction falls to 17% by $A = 25$ and 0% by $A = 30$; Langevin fails already by $A = 15$ (0%, having succeeded at $A = 5$). CBO and PSO fail to reach the target at every depth gap tested here, now confirmed unambiguous under the corrected early-stopping mechanism – which distinguishes a genuine stall from an undisclosed timeout via an independent, algorithm-agnostic stagnation check logged per run (Appendix C.3) – rather than resting on the previous, sentinel-value-conflated "variance collapse" claim.

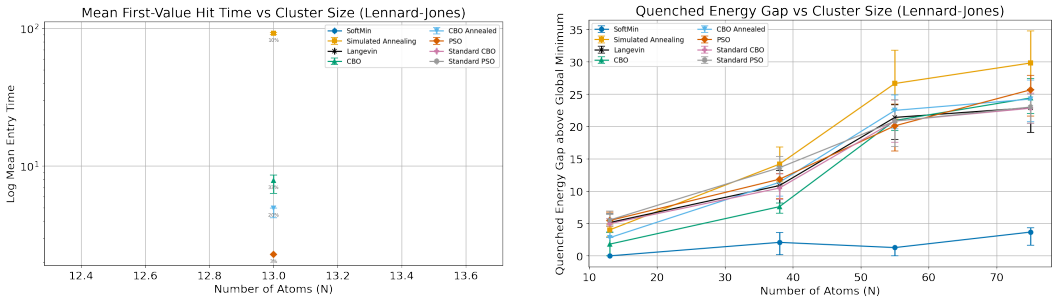


Figure 2: Lennard-Jones cluster minimization, $N \in \{13, 38, 55, 75\}$ atoms ($d = 3N$ up to 225), our scaling-with-dimension benchmark. Left: mean first-value hit time against the strict value-space target; at this threshold almost no method (Softmin included) reliably reaches it within the step budget once $N \geq 38$, so most series simply have no point there – shown honestly rather than suppressed. Right: quenched-energy gap above the literature global minimum (Cambridge Cluster Database, Wales & Doye 1997) of the L-BFGS-B-polished best configuration found by each method, well-defined regardless of whether the hit-time target was reached.

Because the strict hit-time target is rarely reached by any method beyond the smallest cluster size (Figure 2, Left), the quenched-energy gap (Right) is the informative comparison at scale, and it is where Lennard-Jones earns its role as our dimension-scaling benchmark: Softmin's gap stays small and essentially flat across the full range tested (from 0 at $N = 13$ to 3.7 at $N = 75$), while every baseline's gap grows roughly linearly with N, from around 4–5.5 at $N = 13$ to 22–30 at $N = 75$. Softmin's advantage over every baseline *widens*, not narrows, as the cluster grows – the opposite of the ordinary curse-of-dimensionality intuition.

5 Conclusion

In this work, we introduced a novel method for escaping local minima in optimization problems, based on the Softmin Energy interaction dynamics. We proved that, under a Polyak–Łojasiewicz condition, the best objective value across the ensemble decreases at least as fast as ordinary gradient flow on f (Theorem 2.4), for any inverse-temperature schedule: the Softmin coupling never slows down the leading particle. Separately, at fixed temperature, we showed the interaction caps the Freidlin–Wentzell quasipotential barrier between adjacent basins at a level of order $1/\beta$, independently of the true barrier height (Theorem 3.5). Through numerical experiments, we validated our theoretical results and demonstrated the effectiveness of our method on double-well, quadruple-well, and Lennard-Jones cluster benchmarks. In the analyzed benchmarks, our method is the only one that keeps reaching the target as the problem gets harder, outperforming Simulated Annealing, Langevin dynamics, Gradient-Enhanced Particle Swarm Optimization, and Gradient-Aware Consensus-Based Optimization in reliability, and Lennard-Jones quenched-solution quality, even where some baselines remain faster on the easiest settings.

5.1 Limitations and Future Works

Our theoretical analysis provides rigorous guarantees in the practically relevant small-noise regime where the Freidlin–Wentzell Large Deviation Principles apply. While the bound in Theorem 3.5 is

not sharp – in particular u_c and u_h are only upper-estimated by an explicit channel, not computed exactly for general f – a tighter characterization of the true quasipotential, together with intermediate-noise asymptotics, would sharpen the picture. Our equilibrium characterization (Theorem 2.1) requires no convexity assumption on f whatsoever; the ensemble-descent estimate (Theorem 2.4) additionally assumes a Polyak–Łojasiewicz condition on the relevant sublevel set, as discussed after its statement. The exit-time consequence of Theorem 3.5 holds unconditionally only at the level of the quasipotential, and becomes an actual statement about $\mathbb{E}\tau^\sigma$ only under the unverified metastable-domain hypotheses of Corollary 3.6; identifying the exact transition exponent between adjacent basins, and a rigorous large-deviation treatment of the annealed schedule β_t, remain open.

Future work should address several important directions. First, extending convergence proofs and explicit convergence rates to the non-convex regime, particularly for objectives with multiple disconnected basins, would significantly broaden applicability. Second, analyzing the mean-field limit as $n \to \infty$ could provide insights into the scaling of the method with ensemble size and potentially yield dimension-free convergence rates. Third, our experiments (Remark 3.10) found that neither a decaying nor an increasing β_t schedule improved on a well-chosen fixed β on the benchmarks considered, contrary to the original motivation for annealing β over time; understanding whether this is a property of these specific landscapes or a more general phenomenon, and whether an adaptive schedule driven by the ensemble's own state (rather than a fixed function of t) can do better, is an open, concrete direction. Finally, tighter estimates for the exponential rate in the small noise regime can sharpen the theoretical bound to match experimental evidence more closely.

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A Freidlin–Wentzell Large Deviation Theory

This appendix summarizes the background large deviation theory used by Theorem 3.5 and Corollary 3.6. Consider a small-noise gradient diffusion on $\mathbb{R}^N$,

$$\mathrm{d}\mathbf{X}_t = -\nabla U(\mathbf{X}_t)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t,$$

for a potential $U \in C^2(\mathbb{R}^N)$ and standard N-dimensional Brownian motion $(B_t)_{t\geq 0}$; in our application $N = nd$ and $U = nJ_\beta$. As $\sigma \downarrow 0$, sample paths of this diffusion concentrate around trajectories of the noiseless flow $\dot{\mathbf{X}}_t = -\nabla U(\mathbf{X}_t)$, and rare deviations from this flow are governed by the Freidlin–Wentzell large deviation principle [14]. For an absolutely continuous path $\phi : [0, T] \to \mathbb{R}^N$, the Freidlin–Wentzell action is

$$I_{0T}(\phi) := \frac{1}{4}\int_0^T \left\|\dot{\phi}(t) + \nabla U(\phi(t))\right\|^2 \mathrm{d}t, \tag{19}$$

and the quasipotential between a point $x \in \mathbb{R}^N$ and a closed set $A \subset \mathbb{R}^N$ is

$$V(x, A) := \inf_{T>0} \inf_{\substack{\phi(0)=x \\ \phi(T)\in A}} I_{0T}(\phi). \tag{20}$$

Intuitively, $I_{0T}(\phi)$ penalizes deviation from the noiseless flow, vanishing exactly along trajectories of $\dot{\phi} = -\nabla U(\phi)$; $V(x, A)$ is the cheapest such penalty over all paths from x reaching A, and quantifies how atypical (large-deviation-improbable) it is for the diffusion to travel from x to A without the help of noise. The infimum in (20) need not be attained by any single finite-time path: it is generally only approximated arbitrarily well, e.g. by following the time-reversed noiseless flow toward x and then the forward flow toward A, taking arbitrarily long to approach the relevant critical points. This is the sense in which quasipotential values are computed exactly, rather than merely bounded, in the proof of Theorem 3.5 below.

Under the standard hypotheses of Freidlin–Wentzell exit theory [14] – in particular that D is a bounded domain with $x \in D$, that x is the unique asymptotically stable equilibrium of the noiseless flow in D, and that ∂D contains no other equilibria – the exit time τ_D^σ of the diffusion from D, started at x, satisfies

$$\lim_{\sigma\downarrow 0} \sigma \log \mathbb{E}_x \tau_D^\sigma = V(x, \partial D). \tag{21}$$

For a gradient drift, the action of any path admits the exact decomposition

$$I_{0T}(\phi) = U(\phi(T)) - U(\phi(0)) + \frac{1}{4}\int_0^T \left\|\dot{\phi}(t) - \nabla U(\phi(t))\right\|^2 \mathrm{d}t, \tag{22}$$

obtained by expanding the square in (19) and completing it against $\mathrm{d}U(\phi(t))/\mathrm{d}t = \langle \nabla U(\phi(t)), \dot{\phi}(t)\rangle$. Consequently, any segment of ϕ that follows the noiseless flow *downhill* (i.e. $\dot{\phi} = -\nabla U(\phi)$) contributes zero action, while a segment moving *uphill* against the flow (i.e. $\dot{\phi} = \nabla U(\phi)$) costs exactly the potential increase along that segment. This identity is what allows the quasipotential of an explicit admissible path to be computed exactly, rather than merely bounded, in the proof of Theorem 3.5.

B Technical Proofs

Here we will collect some of the more technical proofs of the lemmas stated in the main body.

B.1 Proofs of Section 2

Lemma B.1 (Component gradient and Hessian at critical configurations)**.** *Let* $w_k = e^{-\beta f(x_k)}(\sum_j e^{-\beta f(x_j)})^{-1}$ *denote the Gibbs weight of particle* x_k*, and let* $A_k(\mathbf{x}) = nw_k\left[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))\right]$ *be the dynamic pre-factor. Then*

$$\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_k(\mathbf{x})\nabla f(x_k), \qquad \sum_{k=1}^n A_k(\mathbf{x}) = n. \tag{23}$$

Moreover, for $k \neq \ell$ the mixed Hessian block is the outer product

$$\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_\ell \partial x_k} = \nabla f(x_k)\big(\nabla_{x_\ell} A_k(\mathbf{x})\big)^\top, \tag{24}$$

which vanishes whenever $\nabla f(x_k) = 0$. Consequently, if $\mathbf{x}$ is such that $\nabla f(x_k) = 0$ for every k, then $\nabla(nJ_\beta(\mathbf{x})) = 0$ and the Hessian is block diagonal,

$$\nabla^2(nJ_\beta(\mathbf{x})) = \operatorname{blockdiag}\big(A_1(\mathbf{x})\nabla^2 f(x_1), \ldots, A_n(\mathbf{x})\nabla^2 f(x_n)\big). \tag{25}$$

Finally, if $A_k(\mathbf{x}) = 0$ then $\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_k^2} = -n\beta w_k \nabla f(x_k)\nabla f(x_k)^\top$.

Proof. We compute the derivatives using the chain rule and the properties of the Gibbs weights. First, we establish the gradient of the weight w_i with respect to x_k:

$$\nabla_{x_k} w_i = \begin{cases} \beta w_i w_k \nabla f(x_k) & \text{if } i \neq k, \\ -\beta w_k(1 - w_k)\nabla f(x_k) & \text{if } i = k. \end{cases}$$

For the first derivative of $nJ_\beta(\mathbf{x}) = n\sum_i w_i f(x_i)$, an identical computation to the main text gives $\nabla_{x_k}(nJ_\beta(\mathbf{x})) = nw_k[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))]\nabla f(x_k) = A_k(\mathbf{x})\nabla f(x_k)$; summing over k and using $\sum_k w_k = 1$, $\sum_k w_k f(x_k) = J_\beta(\mathbf{x})$, gives $\sum_k A_k(\mathbf{x}) = n[1 - \beta(J_\beta(\mathbf{x}) - J_\beta(\mathbf{x}))] = n$.

For the mixed second derivative ($k \neq \ell$), we differentiate $\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_k(\mathbf{x})\nabla f(x_k)$ with respect to x_ℓ. Because $\nabla f(x_k)$ is independent of x_ℓ, the Jacobian matrix is the outer product $\nabla f(x_k)(\nabla_{x_\ell} A_k(\mathbf{x}))^\top$. Substituting $\nabla_{x_\ell} w_k = \beta w_k w_\ell \nabla f(x_\ell)$ and $\nabla_{x_\ell} J_\beta(\mathbf{x}) = \frac{1}{n} A_\ell(\mathbf{x})\nabla f(x_\ell)$,

$$\begin{aligned} \nabla_{x_\ell} A_k(\mathbf{x}) &= n(\nabla_{x_\ell} w_k)\left[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))\right] + nw_k\beta\left(\nabla_{x_\ell} J_\beta(\mathbf{x})\right) \\ &= \beta w_\ell\big(nw_k[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))]\big)\nabla f(x_\ell) + \beta w_k A_\ell(\mathbf{x})\nabla f(x_\ell) \\ &= \beta\left[w_\ell A_k(\mathbf{x}) + w_k A_\ell(\mathbf{x})\right]\nabla f(x_\ell), \end{aligned}$$

which gives (24) and vanishes whenever $\nabla f(x_k) = 0$ (or symmetrically $\nabla f(x_\ell) = 0$). If $\nabla f(x_k) = 0$ for every k, every off-diagonal block vanishes and (23) gives $\nabla(nJ_\beta(\mathbf{x})) = 0$; the diagonal block computed below then reduces to $A_k(\mathbf{x})\nabla^2 f(x_k)$, giving the block-diagonal form (25).

Finally, for the diagonal block ($k = \ell$), applying the product rule to $\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_k(\mathbf{x})\nabla f(x_k)$ includes the local Hessian $\nabla^2 f(x_k)$:

$$\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_k^2} = A_k(\mathbf{x})\nabla^2 f(x_k) + \nabla f(x_k)\big(\nabla_{x_k} A_k(\mathbf{x})\big)^\top.$$

Using $\nabla_{x_k} w_k = -\beta w_k(1 - w_k)\nabla f(x_k)$ and $\nabla_{x_k} J_\beta(\mathbf{x}) = \frac{1}{n} A_k(\mathbf{x})\nabla f(x_k)$, a computation identical in structure to the mixed case gives

$$\nabla_{x_k} A_k(\mathbf{x}) = -\beta\left[nw_k + A_k(\mathbf{x}) - 2w_k A_k(\mathbf{x})\right]\nabla f(x_k),$$

so that, substituting back,

$$\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_k^2} = A_k(\mathbf{x})\nabla^2 f(x_k) - \beta\left[nw_k + A_k(\mathbf{x}) - 2w_k A_k(\mathbf{x})\right]\nabla f(x_k)\nabla f(x_k)^\top,$$

which reduces to $-n\beta w_k \nabla f(x_k)\nabla f(x_k)^\top$ when $A_k(\mathbf{x}) = 0$. □

Proof of Theorem 2.1. A locally asymptotically stable equilibrium of the gradient flow (3) is an isolated local minimum of nJ_β: indeed $\frac{d}{dt}(nJ_\beta(\mathbf{x}(t))) = -\|\nabla(nJ_\beta)\|^2 \leq 0$, so if points arbitrarily close to $\bar{\mathbf{x}}$ had strictly smaller value the trajectories starting there could not converge to $\bar{\mathbf{x}}$; and a non-isolated equilibrium admits nearby constant trajectories that likewise do not converge to $\bar{\mathbf{x}}$.

Fix k and freeze the other particles at $\bar{x}_j$, $j \neq k$. Since nJ_β depends on x_k only through $r = f(x_k)$, we may write

$$nJ_\beta(\bar{x}_1, \ldots, x_k, \ldots, \bar{x}_n) = g(f(x_k)), \qquad g(r) := n\,\frac{re^{-\beta r} + C_1}{e^{-\beta r} + C_0},$$

with $C_0 := \sum_{j\neq k} e^{-\beta f(\bar{x}_j)} > 0$ and $C_1 := \sum_{j\neq k} f(\bar{x}_j)e^{-\beta f(\bar{x}_j)}$. By the chain rule and Lemma B.1, wherever $\nabla f(x_k) \neq 0$ one has $g'(f(x_k)) = A_k(\mathbf{x})$; by continuity of both sides in $f(x_k)$, $g'(f(\bar{x}_k)) = A_k(\bar{\mathbf{x}})$. Writing $g'(r) = n\omega(r)[1-\beta(r-g(r)/n)]$ with $\omega(r) = e^{-\beta r}/(e^{-\beta r}+C_0) \in (0,1)$, a direct computation shows that at any critical point r_0 of g (where the bracket vanishes), $g''(r_0) = -n\beta\omega(r_0) < 0$.

Step 1: $A_k(\bar{\mathbf{x}}) \neq 0$. Suppose $A_k(\bar{\mathbf{x}}) = 0$. By the above, $r_0 = f(\bar{x}_k)$ is a critical point of g, and it is a strict local maximum. If f is non-constant near $\bar{x}_k$, there are points y arbitrarily close to $\bar{x}_k$ with $f(y)$ close to but different from r_0, whence $nJ_\beta(\dots, y, \dots) = g(f(y)) < g(r_0) = nJ_\beta(\bar{\mathbf{x}})$, contradicting local minimality. If f is constant near $\bar{x}_k$, moving x_k within that level set leaves every objective value, Gibbs weight, and gradient unchanged, producing a continuum of equilibria arbitrarily close to $\bar{\mathbf{x}}$ and contradicting isolation. Hence $A_k(\bar{\mathbf{x}}) \neq 0$.

Step 2: $\nabla f(\bar{x}_k) = 0$. Since $A_k(\bar{\mathbf{x}}) \neq 0$, $g'(f(\bar{x}_k)) \neq 0$, so g is strictly monotone near $f(\bar{x}_k)$. If $\nabla f(\bar{x}_k) \neq 0$, then f takes values on both sides of $f(\bar{x}_k)$ arbitrarily near $\bar{x}_k$; strict monotonicity of g then yields nearby configurations with $nJ_\beta < nJ_\beta(\bar{\mathbf{x}})$, contradicting local minimality. Hence $\nabla f(\bar{x}_k) = 0$.

Step 3: classification. If $A_k(\bar{\mathbf{x}}) > 0$, then g is strictly increasing near $f(\bar{x}_k)$, so local minimality of nJ_β forces $f(x_k) \geq f(\bar{x}_k)$ for x_k near $\bar{x}_k$; thus $\bar{x}_k$ is a local minimum of f. If $A_k(\bar{\mathbf{x}}) < 0$, the symmetric argument shows $\bar{x}_k$ is a local maximum. Because $nw_k > 0$, the sign of $A_k(\bar{\mathbf{x}})$ is that of $1-\beta(f(\bar{x}_k) - J_\beta(\bar{\mathbf{x}}))$, giving the stated threshold. Finally, $\sum_k A_k(\bar{\mathbf{x}}) = n$ forces some $A_k(\bar{\mathbf{x}}) > 0$, so at least one particle is at a local minimum of f. □

Proof of Theorem 2.4. Fix t and let i attain the minimum, $f(x_i(t)) = m(t)$. Since $f(x_i(t)) \leq f(x_j(t))$ for all j, we have $e^{-\beta_t f(x_j(t))} \leq e^{-\beta_t f(x_i(t))}$, hence $\sum_j e^{-\beta_t f(x_j(t))} \leq ne^{-\beta_t f(x_i(t))}$ and therefore $nw_i(t) \geq 1$. As $J_{\beta_t}(\mathbf{x}(t))$ is a convex combination of the values $f(x_j(t))$, it satisfies $J_{\beta_t}(\mathbf{x}(t)) \geq f(x_i(t))$, so $1-\beta_t(f(x_i(t)) - J_{\beta_t}(\mathbf{x}(t))) \geq 1$. Multiplying,

$$A_i(\mathbf{x}(t)) \geq 1 \qquad \text{for every } i \text{ attaining the minimum at } t. \tag{26}$$

Along such a particle,

$$\frac{d}{dt} f(x_i(t)) = \langle \nabla f(x_i(t)), \dot{x}_i(t)\rangle = -A_i(\mathbf{x}(t))\|\nabla f(x_i(t))\|^2 \leq -\|\nabla f(x_i(t))\|^2.$$

The map $m = \min_k g_k$, with $g_k(t) = f(x_k(t)) \in C^1$, is locally Lipschitz, hence absolutely continuous and differentiable for a.e. t. At a differentiability point t, choose i attaining the minimum; the differentiable function $g_i - m \geq 0$ vanishes at t and thus has zero derivative there, giving $m'(t) = g_i'(t)$. Combined with the previous display,

$$m'(t) = -A_i(\mathbf{x}(t))\|\nabla f(x_i(t))\|^2 \leq -\|\nabla f(x_i(t))\|^2 \leq 0 \qquad \text{for a.e. } t.$$

Integrating, m is nonincreasing, so every minimizing particle stays in $S_0 = \{x : f(x) \leq m(0)\}$, where the Polyak–Łojasiewicz condition (Definition 2.3) applies:

$$m'(t) \leq -\|\nabla f(x_i(t))\|^2 \leq -2\mu\big(f(x_i(t)) - f_\star\big) = -2\mu\big(m(t) - f_\star\big).$$

Setting $y(t) := e^{2\mu t}(m(t) - f_\star)$, which is absolutely continuous, we get $y'(t) = e^{2\mu t}[m'(t) + 2\mu(m(t) - f_\star)] \leq 0$ for a.e. t, so $y(t) \leq y(0)$, i.e.

$$m(t) - f_\star \leq e^{-2\mu t}\big(m(0) - f_\star\big), \qquad t \geq 0. \qquad \square$$

B.2 Proofs of Section 3

Proof of Lemma 3.4. Write $r := r(x)$. Factoring $e^{-\beta f(z_-)}$ out of the Gibbs weights at $\Gamma(x)$,

$$w_1(\Gamma(x)) = \frac{e^{-\beta r}}{n-1+e^{-\beta r}}, \qquad w_j(\Gamma(x)) = \frac{1}{n-1+e^{-\beta r}} \quad (j \geq 2),$$

whence $J_\beta(\Gamma(x)) - f(z_-) = re^{-\beta r}/(n-1+e^{-\beta r}) = u(r)/n$, which is (10). Therefore

$$f(x) - J_\beta(\Gamma(x)) = r - \frac{re^{-\beta r}}{n-1+e^{-\beta r}} = \frac{(n-1)r}{n-1+e^{-\beta r}},$$

and substituting into $A_1(\Gamma(x)) = nw_1(\Gamma(x))[1-\beta(f(x)-J_\beta(\Gamma(x)))]$ gives

$$A_1(\Gamma(x)) = \frac{ne^{-\beta r}}{n-1+e^{-\beta r}} \cdot \frac{n-1+e^{-\beta r}-\beta(n-1)r}{n-1+e^{-\beta r}} = \frac{ne^{-\beta r}q(r)}{(n-1+e^{-\beta r})^2}.$$

Differentiating (10) directly yields the same expression for $u'(r)$, proving (11), consistently with the identity $A_1(\Gamma(x)) = u'(r(x))$ already established in Lemma 3.3. In particular $A_1(\Gamma(x)) = 0$ if and only if $q(r) = 0$, which by the displayed formula for $f(x) - J_\beta$ is equivalent to $f(x) - J_\beta(\Gamma(x)) = 1/\beta$, i.e. to the vanishing of the Softmin threshold gap of the active particle.

Since $q'(r) = -\beta(n-1) - \beta e^{-\beta r} < 0$ for all $r \in \mathbb{R}$, $q(0) = n > 0$, and $q(r) \to -\infty$ as $r \to \infty$, q has a unique zero $r_c > 0$. From $q(1/\beta) = e^{-1} > 0$ we get $r_c > 1/\beta$; and $q(r_c) = 0$ reads $(n-1)(\beta r_c - 1) = e^{-\beta r_c} < 1$, giving $\beta r_c < n/(n-1)$. This is (12). As $ne^{-\beta r}/(n-1+e^{-\beta r})^2 > 0$, the sign of u' is the sign of q, so u is strictly increasing on $(-\infty, r_c]$ and strictly decreasing on $[r_c, \infty)$. □

Proof of Theorem 3.5. Part 1. For $r \le 0$ we have $u(r) \le 0$. For $r > 0$, since $n-1+e^{-\beta r} > n-1$,

$$u(r) < \frac{n}{n-1} r e^{-\beta r} \le \frac{n}{n-1} \max_{\rho \ge 0} \rho e^{-\beta\rho} = \frac{n}{e(n-1)\beta},$$

the maximum being attained at $\rho = 1/\beta$. Hence $u_c = \max_{\mathbb{R}} u < n/(e(n-1)\beta)$, which is (14); the bound on r_c is (12).

Part 2. A channel path has the form $\phi(t) = \Gamma(\psi(t))$; the frozen coordinates stay at the critical point z_-, so their drift vanishes and, with $W(x) := u(r(x))$, the action reduces to $I_{0T} = \frac{1}{4}\int_0^T |\dot\psi + W'(\psi)|^2 \mathrm{d}t$. The one-dimensional version of (22) is

$$I_{0T} = W(\psi(T)) - W(\psi(0)) + \frac{1}{4}\int_0^T |\dot\psi - W'(\psi)|^2 \mathrm{d}t.$$

By (7), r is a strictly increasing bijection from $[z_-, s]$ onto $[0, h]$, so there is a unique $x_m \in (z_-, s]$ with $r(x_m) = \min\{h, r_c\}$. On $[z_-, x_m]$ we have $r(x) \le r_c$, so by Lemma 3.4 W is strictly increasing there, with total rise $W(x_m) - W(z_-) = u_h$; on $[x_m, s]$ we have $r(x) \ge r_c$, so W is nonincreasing. Every continuous channel path from z_- to s passes through x_m, and applying the displayed decomposition to the sub-path up to the first hitting time of x_m bounds the action from below by u_h. This value is attained in the quasipotential sense by following the reversed deterministic flow from z_- up to x_m (cost exactly u_h) and then the forward flow toward s at zero additional cost, with the standard finite-time approximations near the critical points. Therefore the channel quasipotential equals u_h, and since V_β is an infimum over the larger class of all paths in $\mathbb{R}^n$,

$$V_\beta(\Gamma(z_-), \Gamma(s)) \le u_h.$$

Because $u(r)/r = ne^{-\beta r}/(n-1+e^{-\beta r}) < 1$ for $r > 0$, we have $u_h < \min\{h, r_c\} \le h$, which gives (15).

Part 3. Assume $r_c < h$. At $\Gamma(s)$ every particle sits at a critical point of f (either s or z_-). For the active particle, (11) gives $A_1(\Gamma(s)) = u'(h) < 0$ because $h > r_c$, and $f''(s) < 0$ since s is a nondegenerate maximum. For a frozen particle at z_-,

$$A_j(\Gamma(s)) = nw_j(\Gamma(s))\big[1 + \beta\big(J_\beta(\Gamma(s)) - f(z_-)\big)\big] > 0$$

by (10), since $J_\beta(\Gamma(s)) - f(z_-) = u(h)/n > 0$, and $f''(z_-) > 0$. Hence Corollary 2.2 applies and $\Gamma(s)$ is a strict local minimum of nJ_β. On the z_- side, r decreases strictly from h through r_c as x moves from s toward z_- (recall $x_m \in (z_-, s)$ is where $r(x_m) = r_c$, well-defined since $r_c < h$), so the channel well at $\Gamma(s)$ has barrier $u_c - u(h)$ on this side unconditionally — this uses only $r_c < h$, not ρ_+.

The z_+ side depends on ρ_+. On (s, z_+), r decreases strictly from h to ρ_+ by (7)–(8).

If $\rho_+ < r_c$, then r crosses r_c on this interval, so there is a unique ξ_+ with $r(\xi_+) = r_c$. Between s and ξ_+ the reduced potential rises from $u(h)$ to u_c; the gradient-action argument of Part 2 gives channel action $u_c - u(h)$, and since $0 < u(h) < u_c$ we obtain (17), so the well has matching barrier heights on both sides. Finally, for $x \in (\xi_+, z_+)$ we have $r(x) < r_c$, hence $A_1(\Gamma(x)) = u'(r(x)) > 0$

by (11), while $f'(x) < 0$; therefore the active coordinate of the deterministic channel flow satisfies $\dot{x} = -A_1(\Gamma(x))f'(x) > 0$, driving the particle toward z_+. Concatenating the admissible path of Part 2 (from $\Gamma(z_-)$ to $\Gamma(s)$, action u_h) with this one (from $\Gamma(s)$ to $\Gamma(\xi_+)$, action $u_c - u(h)$) gives an admissible path from $\Gamma(z_-)$ to $\Gamma(\xi_+)$ of total action $u_h + (u_c - u(h))$, proving (18).

If instead $\rho_+ \geq r_c$, then $r(x) \in [r_c, h)$ for every $x \in (s, z_+)$: the interval is entirely on the decreasing side of u, where u and r are both strictly decreasing in x, so $u(r(x))$ is strictly increasing in x throughout (s, z_+), from $u(h)$ to $u(\rho_+)$. No interior peak occurs, so no point ξ_+ with $r(\xi_+) = r_c$ exists in (s, z_+), and the channel presents no barrier separating $\Gamma(s)$ from $\Gamma(z_+)$ on this side.

If instead $h \leq r_c$, then $r(x) \leq h \leq r_c$ for x near s, with equality only at $x = s$, and the strict monotonicity of u on $(-\infty, r_c]$ shows that $x \mapsto u(r(x))$ has a strict local maximum at s. Hence $\Gamma(s)$ is not a local minimum of nJ_β. □

C Appendix: Experimental Details and Hyperparameters

To ensure full reproducibility of the numerical experiments presented in Section 4, this appendix details the hyperparameters and initialization schemes used across all benchmark landscapes. All continuous-time dynamics were discretized using the Euler-Maruyama method. The source code used to execute these experiments is publicly available at [3].

C.1 Hardware and Computational Resources

All numerical experiments were executed on the *Toeplitz* Scientific Computing Cluster hosted by the Department of Mathematics at the University of Pisa. To fully leverage the vectorized nature of the simulations and evaluate independent trials in parallel, jobs were strictly allocated to a single compute node at a time. The cluster nodes utilized for these experiments include:

- **High-Capacity Nodes:** AMD EPYC 7763 processors featuring 128 physical cores (64 cores per socket, 2 sockets, 256 logical threads) and 2 TB of RAM.
- **Standard Nodes:** Intel Xeon CPU E5-2650 v4 @ 2.20GHz featuring 24 physical cores (12 cores per socket, 2 sockets) and 256 GB of RAM. These nodes were specifically utilized to parallelize the independent runs of the experiments using Python's `multiprocessing` framework.

C.2 General Experimental Settings

Table 1 summarizes the shared parameters utilized across all algorithms for each specific landscape. The swarm size and maximum step limits were kept constant to provide a fair baseline for evaluating the topological escape mechanisms. Domain bounds for the 1D and 2D wells are derived directly from each landscape's own critical points rather than set by hand: the domain extends $K = 5$ natural harmonic fluctuation-lengthscales ($1/\sqrt{f''}$ at the relevant minimum, or the softest Hessian eigendirection in 2D) beyond the outermost minimum – the same role a 5σ truncation plays for a locally-Gaussian process, giving negligible boundary-contamination probability without an oversized domain. A fixed absolute margin was checked and found to represent anywhere from ~ 1 to ~ 5.5 lengthscales across the swept difficulty range, an uncalibrated $> 5\times$ inconsistency this formula removes. The quadruple well is swept over $a \in \{2, \ldots, 30\}$ (not 1): at $a = 1$ the softest Hessian eigenvalue at the well vanishes exactly, so the landscape does not yet have a genuine local minimum in that direction.

C.3 Hyperparameter Selection Methodology

Each of the 100 independent runs (Table 1) is indexed by an integer seed $\in \{0, \ldots, 99\}$, and every worker seeds NumPy's global RNG once via `np.random.seed(seed*1000)` before drawing that run's initial swarm and simulating every algorithm on it. This seed indexing is reused identically at every sweep point (every δ, A, or d) rather than redrawn independently per point: `seed=7` at $\delta = 5$ and `seed=7` at $\delta = 25$ consume the same underlying draw from the same seeded RNG state, remapped only by that point's own initialization bounds. Consequently the 100-run samples at different sweep points are correlated, not independent, and the quartile error bars in Figure 1 should

Table 1: General settings and landscape configurations.

Parameter	Value / Description
Swarm Size (n)	100
Independent Runs	100 (1D/2D wells, Ackley), 30 (Lennard-Jones)
Max Step Limit	10^7 (1D/2D wells, Ackley), 10^5 (Lennard-Jones)
1D Double Well	$f(x) = x^4 - ax^2 + 0.05x$ (asymmetric: the linear tilt breaks the left/right symmetry, giving a genuinely suboptimal starting basin – see §C.3)
Initialization	Uniformly sampled around the shallower (suboptimal) minimum z_-
Domain Bounds	$[z_- - 5/\sqrt{f''(z_-)},\ z_+ + 5/\sqrt{f''(z_+)}]$, $f''(x) = 12x^2 - 2a$ ($z_\pm$: the two minima)
2D Quadruple Well	$f(x, y) = x^4 + y^4 - a(x^2 + y^2) + xy$
Initialization	Uniformly sampled around the shallower diagonal minimum (r_-, r_-) (genuinely suboptimal starting basin); target is the deeper anti-diagonal minimum $(r_+, -r_+)$ – see §C.3
Domain Bounds	$[-(r_+ + m),\ r_+ + m]^2$, $r_- = \frac{1}{2}\sqrt{2a-1}$, $r_+ = \frac{1}{2}\sqrt{2a+1}$ ($\geq r_-$), $m = 5/\sqrt{\lambda_{\min}}$, $\lambda_{\min} = \min\big(f_{xx}(r_-, r_-) - \lvert f_{xy}(r_-, r_-)\rvert,\ f_{xx}(r_+, -r_+) - \lvert f_{xy}(r_+, -r_+)\rvert\big)$
High-Dimensional Ackley	Generalized Ackley ($d \in [10, 90]$)
Function Parameters	$a = 20.0$, $b = 0.2$, $c = 2\pi$
Initialization	Uniformly sampled in $[-10, 10]^d$
Domain Bounds	$[-10, 10]^d$
Lennard-Jones Cluster	$E(\mathbf{x}) = \sum_{i<j} 4(r_{ij}^{-12} - r_{ij}^{-6}) + E_{\text{conf}}$, $N \in \{13, 38, 55, 75\}$ atoms ($d = 3N$)
Initialization	Uniform-in-ball, rejecting pairs closer than 0.9
Domain Bounds	$[-(R_c + 2),\ R_c + 2]^d$, $R_c = 1.5N^{1/3} + 1.5$

be interpreted as within-seed variability across the ensemble rather than as independent per-point confidence intervals.

Learning rates are unified to $\eta = 0.001$ across every benchmark and algorithm (the Theorem 3.5 verification drivers excepted, which use their own, separately-motivated small-noise/small-step-size regime). The Softmin inverse temperature β is selected per benchmark family via a two-stage, code-audited procedure (`src/hyperparameter_diagnostics.py`), evaluated at a representative *middle*-of-sweep setting rather than the hardest one, so the choice is not fit to the one condition the sweep is meant to stress-test: (1) an initial screen measures the distribution of $\beta(f_i - J_\beta)$ across the ensemble, flagging candidate values where trailing particles are not uniformly driven into the exponentially-suppressed regime; (2) because that screen alone favors values that are too small – it only measures whether every particle receives rescue-drift, not what the reported first-entry-time metric rewards, namely the single fastest particle reaching the target, which benefits from a *concentrated* ensemble via the leading particle's prefactor approaching n – the candidates are then validated directly by measuring actual entry time. Across every benchmark tested (1D double well, 2D quadruple well, high-dimensional Ackley, and a spot check on Lennard-Jones), this second step confirmed $\beta = 10$ as at or near the empirical optimum, matching the value used throughout; going lower is consistently slower or fails to reach the target at all, and going higher does not help (and is markedly worse on the quadruple well). The same direct-validation procedure was applied to Simulated Annealing's geometric cooling rate, and to comparing β held fixed against both a decaying and an increasing schedule – see Remark 3.10 for the latter result. The Lennard-Jones benchmark additionally clips the gradient's Lennard-Jones term (never the confinement term, whose restoring force must scale freely to remain able to pull back an atom that has drifted outside the container) to a per-atom norm of 1.0 before the swarm's discrete update, applied identically to every algorithm; this clip is not applied to the L-BFGS-B local polish ("quench") of each method's final configuration,

which uses the true, unclipped gradient throughout, since a quasi-Newton method's curvature estimate is only valid for the true gradient.

Baselines received the same budget, not literature defaults asserted without checking. Simulated Annealing, CBO, and PSO were given the identical diagnostic-guided procedure and, where a comparable parameterization exists, the identical candidate budget as Softmin's β (Table 2), rather than unexamined literature-standard values. *Simulated Annealing*: seven candidate cooling rates were derived from target cooling *timescales* $t^\star \in \{10^3, 3{\times}10^3, 10^4, 3{\times}10^4, 10^5, 3{\times}10^5, 10^6\}$ (steps to reach 5% of the initial temperature, rate $= 0.05^{1/t^\star}$) rather than hand-picked decimals, and validated directly the same way as β; the shipped rate (0.99995) was reconfirmed on both wells. SA's initial temperature T_0 (previously an unexamined a priori default of 10 – not a literature-sourced value; T_0 is inherently problem-scale-dependent, so no universal literature default exists) was calibrated separately, via a Kramers/Arrhenius criterion rather than escape time (minimizing escape time is degenerate for T_0 in this continuous, always-accept diffusion: larger T_0 is monotonically faster with no interior optimum, eventually indistinguishable from bounded random search): five candidates $T_0 = f \cdot h$, $f \in \{0.25, 0.5, 1, 2, 4\}$, where h is the double well's mid-sweep barrier height, were scored by the closed-form escape factor $\exp(-h/T_0)$ landing in a *low* target band $[0.01, 0.05]$ – a genuine, non-trivial barrier, not the near-unity band a literal reading of the discrete-Metropolis initial-acceptance-ratio heuristic (Kirkpatrick et al., 1983) would suggest, which would leave the barrier already thermodynamically negligible from $t = 0$ and trivialize exactly the obstacle this baseline is meant to face. $T_0 = h/4$ was selected and validated directly, reaching the target within a few hundred steps at the calibration midpoint and transferring to the quadruple well; like CBO and PSO below, it fails at the hardest points of the double-well sweep ($\delta \gtrsim 25$), reported as a genuine negative result rather than papered over with a per-δ schedule – independently re-confirmed (2026-08-07) with early-stopping disabled entirely and a 1,500,000-step horizon (15× the original probe): `best_global_value` is bit-for-bit identical from step 10,000 onward at both $\delta = 25$ and $\delta = 30$, consistent with SA's own geometric cooling schedule driving the temperature to ≈ 0 well before step 200,000 and permanently freezing the dynamics, not a premature-abort artifact. The same criterion applied to Ackley's own barrier scale was inconclusive even at a five-times-longer diagnostic budget, so Ackley and Lennard-Jones retain $T_0 = 10$. *CBO*: the same 11-value β grid used for Softmin was swept, together with two annealed-schedule variants (growth-only, unlike Softmin's decaying schedule, matching CBO's own literature convention). The result is a genuine negative finding, not an oversight: every candidate times out via consensus collapse on the double well, the quadruple well, *and* Ackley, at the identical step *regardless of which β was used* – direct testing across $\beta \in \{0.01, 0.02, 0.1, 1.0, 10.0\}$ found position_collapse at the same step on every benchmark tested, confirming the collapse is a β-independent structural property of vanilla CBO's self-shrinking anisotropic diffusion, not a symptom of a poorly-chosen β. Since the choice is empirically immaterial, $\beta = 0.01$ (CBO's literature-typical range; unlike PSO's constriction coefficients below, CBO has no single canonical numeric β, but 0.01–0.02 is the commonly cited scale) is used in place of the previous unexamined a priori default of 10. *PSO* has no single temperature-like scalar to sweep, so six named, literature-grounded (w, c_1, c_2) triples were tested instead (the canonical Clerc–Kennedy constriction coefficients used throughout; the original Kennedy–Eberhart rule; both endpoints of the Eberhart–Shi linearly-decreasing-inertia schedule; a social-weighted variant biased toward the swarm's global best; and a deliberately non-contractive stress-test triple), pre-filtered by a finite/at-bound stability probe. On the double well and quadruple well, PSO shows the same negative result as CBO. On Ackley, however, the literature-standard default itself *fails* – timing out via consensus collapse at every dimension tested ($d = 50, 70, 90$) – while two of the six alternatives succeed at all three; the faster, $(w, c_1, c_2) = (0.4, 2.0, 2.0)$, was adopted for Ackley and is used for the Ackley-d=50 ablation in §C.4.4; a spot check found it also improves the Lennard-Jones quenched result, so it is used there too (Table 1).

Table 2: Calibration budget: candidate values/triples tested, per benchmark family, for each algorithm's diagnostic-guided tuning pass. PSO's lower total reflects having no schedule mechanism to test (see text), not less scrutiny. SA's "schedule variants" column is its T_0 candidates, evaluated once at the double-well midpoint (not per benchmark, see text) rather than a decaying/growing schedule.

Algorithm	**Fixed candidates**	**Schedule/T_0 variants**	**Total per benchmark**
Softmin	11	2	13
SA	7	5	12
CBO	11	2	13
PSO	6	0	6

Table 3: Hyperparameters for Softmin Energy Dynamics.

Parameter	**Value**
Learning Rate (η)	0.001 (Benchmarks), 0.01 (Thm. 3.5 Verification)
Adaptive Learning Rate (Adagrad)	Disabled (False)
Noise Intensity (σ)	1.0 (Benchmarks), 0.05 (Thm. 3.5 Verification)
Noise Distribution	Normal
Fixed Inverse Temperature (β)	10.0 (Benchmarks), 2.0 (Thm. 3.5 Verification)

Table 4: Hyperparameters for Simulated Annealing (SA). T_0 differs by benchmark family: the wells' value is Kramers-factor calibrated and validated (see §C.3); Ackley/Lennard-Jones retain the unexamined a priori default of 10 (not a literature value – T_0 has no problem-independent literature standard), since the same calibration attempt there was inconclusive.

Parameter	**Value**
Learning Rate (η)	0.001
Base Noise Variance Factor	$2 \times \sigma \times \eta$
Initial Temperature (T_0)	14.0283 (1D/2D wells), 10.0 (Ackley, Lennard-Jones)
Cooling Schedule	Geometric
Geometric Cooling Rate	0.99995 (1D/2D wells, re-validated via a 7-candidate timescale sweep), 0.9999 (Ackley, Lennard-Jones)

Table 5: Hyperparameters for Gradient-Aware Particle Swarm Optimization (PSO). The literature-standard constriction coefficients were found, by direct escape-time testing, to fail on Ackley specifically (all dimensions tested); a diagnostic-guided alternative is used there instead (see §C.3).

Parameter	**Value**
Gradient Step Learning Rate (η)	0.001
Inertia Weight (w)	0.729 (1D/2D wells), 0.4 (Ackley, Lennard-Jones)
Cognitive Coefficient (c_1)	1.49445 (1D/2D wells), 2.0 (Ackley, Lennard-Jones)
Social Coefficient (c_2)	1.49445 (1D/2D wells), 2.0 (Ackley, Lennard-Jones)

Table 6: Hyperparameters for Consensus-Based Optimization (CBO). $\beta = 0.01$, CBO's literature-typical range (unlike PSO's constriction coefficients below, CBO has no single canonical numeric β, but 0.01–0.02 is the commonly cited scale) – switched from an unexamined a priori default of 10 after directly confirming the switch changes nothing: an 11-value sweep including $\beta = 0.01$ plus two annealed-schedule variants found consensus collapse at the identical step regardless of β, on all three main benchmarks (a genuine, β-independent structural negative finding, see §C.3).

Parameter	**Value**
Learning Rate (Δt)	0.001
Consensus Weight (λ)	1.0
Exploration Noise (σ_{CBO})	$\sqrt{2}$
Inverse Temperature (β)	0.01

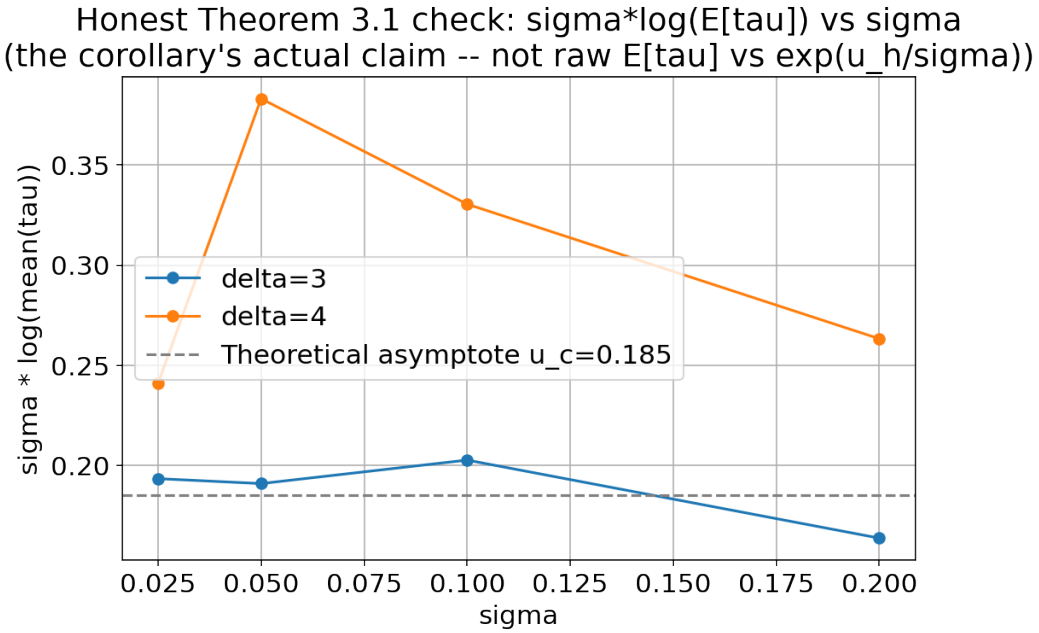


Figure 3: $\sigma \log(\text{mean } \tau)$ vs. σ at $\delta = 3, 4$ (both already in the saturated $h > r_c$ regime), against the theoretical asymptote $u_c = u(r_c, \beta, n)$. This is the quantity Theorem 3.5's corollary actually constrains – $\lim_{\sigma \to 0} \sigma \log \mathbb{E}[\tau] = V_\beta \leq u_c$ – unlike a direct overlay of raw $\mathbb{E}[\tau]$ against $\exp(u_c/\sigma)$ at one fixed, non-vanishing σ, which conflates the proven log-asymptotic rate with an uncontrolled, h-dependent sub-exponential prefactor. Both series trend toward u_c as σ shrinks, consistent with the corollary; neither is monotonic all the way down, which is expected finite-σ behavior, not a violation.

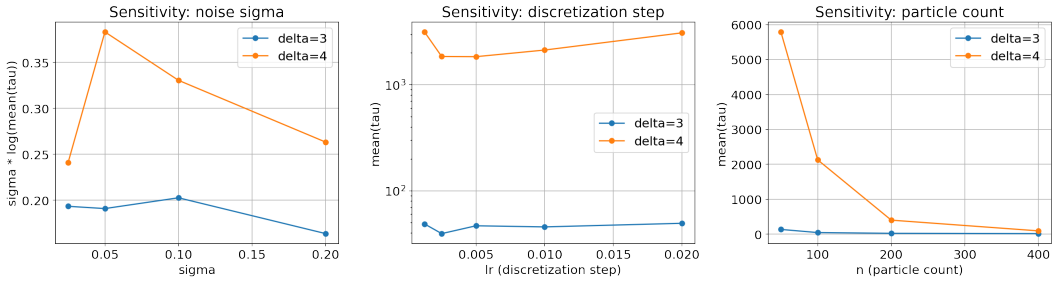


Figure 4: Sensitivity of the same corollary check to the discretization step η and particle count n, at $\delta = 3, 4$. $\sigma \log(\text{mean } \tau)$ (left) reproduces Figure 3; mean τ vs. η (center) should flatten as $\eta \to 0$ if the discretization is close to its continuous-time limit; mean τ vs. n (right) should only ever improve with more particles.

Table 7: Mean per-particle gradient/objective evaluation counts and wall-clock time, averaged over the full double-well/quadruple-well sweep actually run (not a short synthetic probe). Counts are per-particle (n_particles $\times$ steps), not per vectorized call. Because a run that never escapes contributes its full step budget to the average, SA's and Langevin's large counts partly reflect how often they exhaust the budget without reaching the target (Figure 1), not a per-step cost difference – every gradient-based method performs the same $O(n)$ work per step. Standard CBO/PSO have zero gradient evaluations by construction (gradient-free); Standard CBO's nonzero objective-evaluation count above PSO's reflects its consensus-weight computation, which re-evaluates the objective every step.

Method	**Gradient Evals**	**Objective Evals**	**Wall-clock (s)**
SoftMin	502,150	502,150	11.68
Simulated Annealing	14,149,330	14,149,330	351.57
Langevin	42,907,470	42,907,470	1033.74
CBO	110,514	110,514	2.79
CBO Annealed	110,807	110,807	2.79
PSO	99,868	99,868	2.43
Standard CBO	0	1,363,282	18.08
Standard PSO	0	99,852	1.31

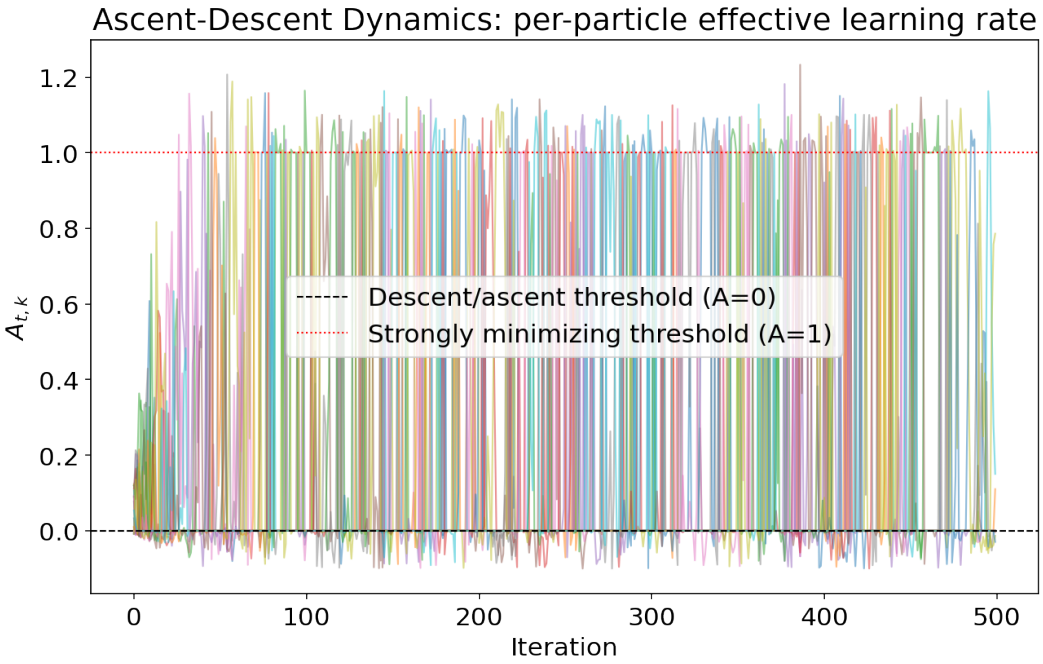


Figure 5: Per-particle effective learning rate $A_k(\mathbf{x}_t)$ over a single Softmin run on the 1D double well ($\delta = 4$, 20 particles). Particles with $A_k(\mathbf{x}_t) > 0$ descend (more strongly so above the $A_k(\mathbf{x}_t) = 1$ line); particles with $A_k(\mathbf{x}_t) < 0$ ascend, the mechanism the ablation study below tests directly.

C.4 Further Experimental Results

C.4.1 Theory Verification

C.4.2 Computational Cost Comparison

C.4.3 Ascent-Descent Mechanism Illustration

C.4.4 Ablation Study: Ascent-Descent Mechanism

We isolate the contribution of the sign-changing prefactor $A_k(\mathbf{x})$ itself by comparing the real (*signed*) mechanism against two ablations: *positive part*, $(A_k(\mathbf{x}))_+$, which freezes any particle the real mechanism would have pushed uphill instead of letting it ascend; and *absolute value*, $|A_k(\mathbf{x})|$, which forces every particle to always descend, at whatever magnitude the real mechanism would have applied. All three variants share the same Gibbs/softmax weighting $n \cdot w_k$ that scales $A_k(\mathbf{x})$ in the first place – only the sign treatment on top of it differs.

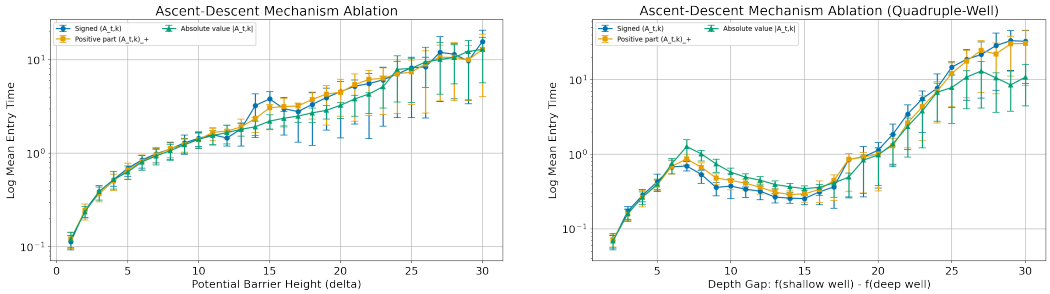


Figure 6: Ascent-descent mechanism ablation on the two swept well benchmarks (mean entry time $\pm$ IQR vs. difficulty, same value-space target and status-based masking as Figure 1).

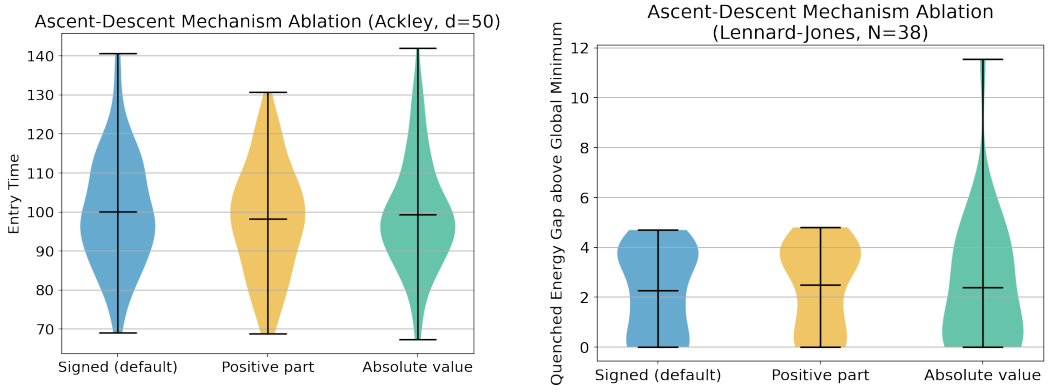


Figure 7: Ascent-descent mechanism ablation at two fixed, hard settings with no natural sweep axis (Ackley $d = 50$, 100 seeds; Lennard-Jones $N = 38$, 30 seeds), shown as per-seed distributions (violin plots) rather than a single mean $\pm$ IQR point per mode, since the three modes turn out to substantially overlap here.

The result is not the clean "signed wins, the ablations degrade" story the sign-flipping mechanism might suggest. On the two swept well benchmarks (Figure 6), there is no consistent winner: at the easier settings signed is competitive or best (e.g. quadruple well $A = 15$: signed 0.25, positive part 0.29, absolute 0.34), but at harder settings absolute is clearly faster – double well $\delta = 15$: signed 3.81, positive part 3.06, absolute 2.20; quadruple well $A = 30$: signed 32.69, positive part 30.70, absolute 10.82, roughly $3\times$ faster than either alternative that still preserves or freezes the sign. At the two fixed, hard settings (Figure 7) the three modes are close to indistinguishable: Ackley $d = 50$ entry

time (mean 100.1/98.2/99.4 for signed/positive part/absolute, all 100% reached) and Lennard-Jones $N = 38$ quenched-energy gap (2.26/2.48/2.40) show heavily overlapping distributions.

Taken together, these four benchmarks suggest that preserving the sign of $A_k(\mathbf{x})$ – letting trailing particles genuinely ascend rather than merely stall or be redirected downhill – is not what is doing most of the practical work in this noise regime ($\sigma = 1$). The Gibbs/softmax weighting $n \cdot w_k$, identical across all three variants, already strongly suppresses trailing particles' contribution regardless of the sign convention applied on top of it; that shared attenuation is the more plausible source of the empirical advantage over the baselines in Section 4. The sign-flipping ascent mechanism remains what Theorem 3.5 is stated and proved for, and is a $\sigma \to 0$ asymptotic statement – these benchmarks run at a fixed, non-vanishing σ, and the ablation results here are consistent with the sign convention mattering more as σ shrinks than it does at the noise level tested.